

CRYPTO BOT PROJECT CONTEXT

Auto-Trading Bot untuk Altcoin dengan Momentum Derivative Signals

1. STRATEGY OVERVIEW

Core Philosophy

- **Signal:** Momentum derivative (acceleration-based), NOT lagging indicators (RSI/MACD)
- **Timeframe:** Multi-TF bias filter (4H for bias, 1H for entry)
- **Asset:** Crypto altcoins only (top 20-30 by market cap, >\$1B cap, >\$50M 24h volume)
- **Execution:** Aggressive entry on small TF momentum, conservative exit

Trade Flow

4H Bias Filter (Direction):

- └ ROC(close, 9) trend check
- └ Volume confirmation
- └ Set bias: UP/DOWN/NONE

↓

1H Momentum Trigger (Entry):

- └ momentum = close - close[5]
- └ acceleration = momentum - momentum[1]
- └ Check: acceleration_inflection + momentum_direction + volume spike
- └ IF all align with 4H bias → EXECUTE

↓

Position Management:

- └ TP: 2-3% (1H move realistic for alts)
 - └ SL: ATR(14) * 1.5 (dynamic)
 - └ Time exit: 4-6 candles (don't hold beyond small window)
 - └ Early close: If acceleration collapse (protect capital)
-

2. SIGNAL DEFINITIONS (Detailed)

Momentum Derivative Calculation

1H Entry Signal:

```
python

# Core momentum
momentum_1h = close[now] - close[5_bars_ago]

# Acceleration (derivative of momentum)
acceleration_1h = momentum_1h - momentum_1h[1_bar_ago]

# Signal conditions
is_accel_building = acceleration_1h > previous_acceleration_1h[1]
is_momentum_positive = momentum_1h > 0
is_momentum_negative = momentum_1h < 0

# Volume confirmation
volume_spike = volume_1h > SMA(volume, 20) * 1.2
```

4H Bias Filter:

```
python

# Trend direction (long-term bias)
roc_4h = (close[now] - close[20_bars_ago]) / close[20_bars_ago]

# Conditions
bias_up = roc_4h > 0 AND volume_avg_4h increasing
bias_down = roc_4h < 0 AND volume_avg_4h increasing
bias_none = abs(roc_4h) < threshold OR low_volume

# Apply
only_BUY_if bias_up OR bias_neutral
only_SELL_if bias_down OR bias_neutral
SKIP_all if bias_none (range market)
```

Entry Rules (ALL must be TRUE)

BUY Setup:

- ☐ 4H bias is UP or NEUTRAL (not strong DOWN)
- ☐ 1H acceleration positive (building momentum)

- ☐ 1H momentum positive
- ☐ 1H volume > 1.2x 20-period average
- ☐ Price NOT at liquidation level (if using Coinglass)
- ☐ Funding rate reasonable (if using Coinglass)

SELL Setup:

- ☐ 4H bias is DOWN or NEUTRAL (not strong UP)
- ☐ 1H acceleration negative (collapsing momentum)
- ☐ 1H momentum negative
- ☐ 1H volume > 1.2x 20-period average
- ☐ Same filters as BUY

Exit Rules (ANY trigger closes position)

Take Profit:

- Price hit +2-3% from entry
- Hold 4-6 candles max (if no TP hit, reassess)

Stop Loss:

- Price hit $-1.5 \times \text{ATR}(14)$ from entry
- OR acceleration collapses ($\text{momentum_derivative} < 0$)
- OR time expires without TP

3. COIN SELECTION

Criteria

- **Market Cap:** > \$1B
- **24h Volume:** > \$50M
- **Liquidity:** Bid-ask spread < 0.15% (on Binance)
- **Exclude:** Meme coins, leverage coins, low-cap speculation

Recommended Watchlist (Update monthly)

Tier 1 (Most Liquid):
 - SOL, AVAX, ADA, LINK, DOGE, XRP

Tier 2 (Good Liquidity):

- ARB, OP, NEAR, POLYGON, ATOM

Tier 3 (Volatile but tradeable):

- INJ, SEI, SUI, APTOS, JUP

Position Limits

- Max 5-10 simultaneous positions (avoid correlation trap)
 - Each position = 1-2% portfolio risk
 - No more than 3 trades from same sector (e.g., Layer2 ecosystem)
-

4. API & DATA SETUP

Binance API (FREE)

What you get:

- Real-time OHLCV data (all TFs)
- Funding rates
- Order book
- Trade execution
- Account info
- Cost: \$0 (pay only trading fees)

Rate Limits: 1,200 req/min (enough for bot)

Setup:

1. Go binance.com → Account → API Management
2. Create API Key (enable "Full Access" except "Enable Withdrawals")
3. IP whitelist your bot IP (or leave empty for testing)
4. Copy API Key + Secret Key
5. Store securely (env variables, not hardcoded)

Python Library:

```
bash
```

```
pip install python-binance
```

Coinglass API (OPTIONAL - Free tier or \$29/month)

What you get (Free 10k calls/month):

- Real-time liquidation data
- Cross-exchange long/short ratios
- Funding rate comparison
- Open interest trends

Use case: Risk filtering (avoid liquidation traps)

Setup:

1. Go `coinglass.com` → Register
2. Create API key in dashboard
3. 10,000 calls/month free
4. Paid: Hobbyist \$29/month for 500k calls

When to add: After first liquidation loss or if want risk reduction

5. BOT ARCHITECTURE

Tech Stack (Recommended)

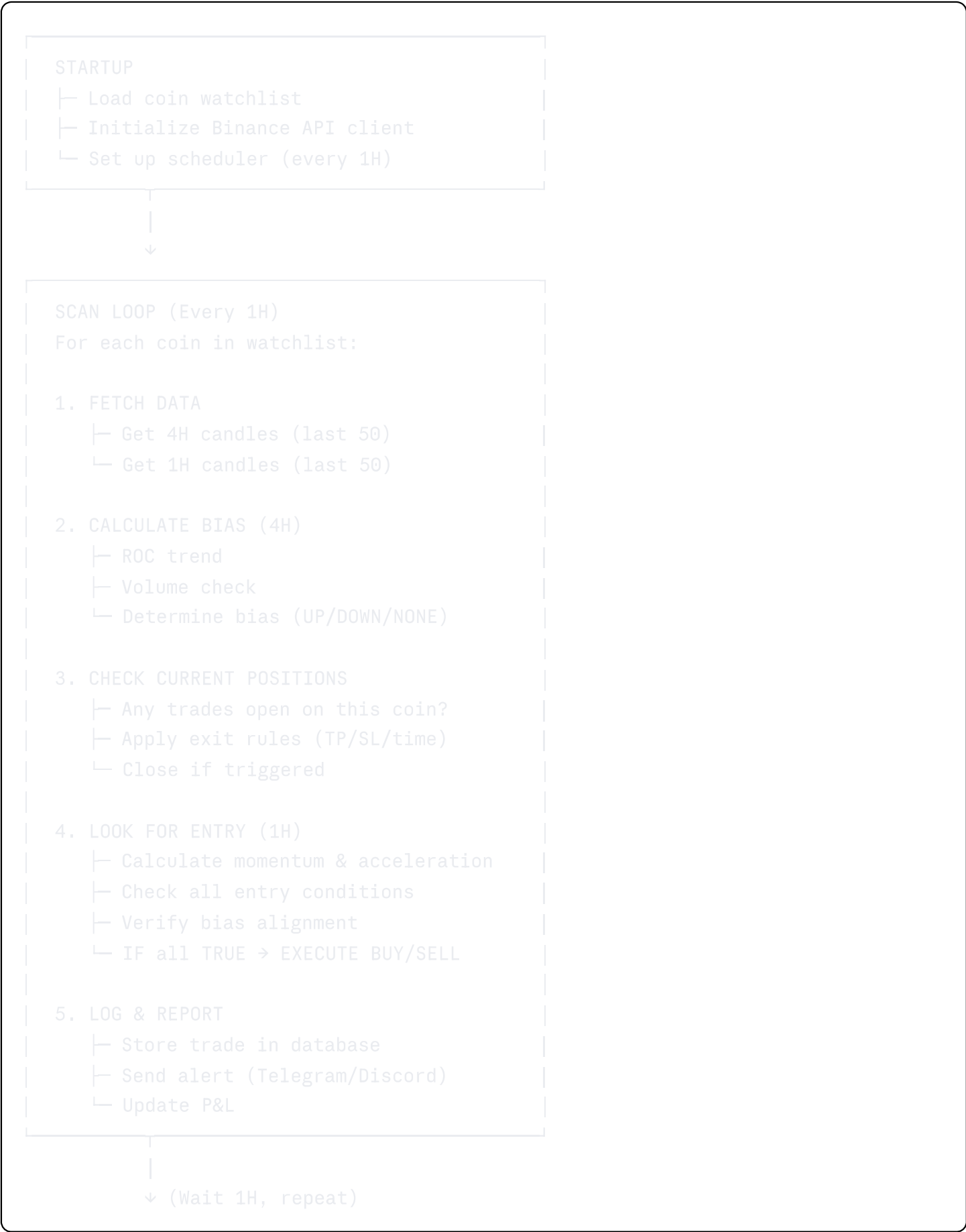
Option A: Python (Best for learning + flexibility)

- Language: Python 3.9+
- Binance lib: `python-binance`
- Backtest: `backtrader` or `zipline`
- Scheduler: `APScheduler`
- Data: Local JSON or database
- Execution: Direct Binance API

Option B: Node.js (Faster, better for production)

- Language: Node.js
- Binance lib: `binance-api-node`
- Backtest: `TradingBots` or custom
- Scheduler: `node-cron`
- Execution: Direct Binance API

Bot Flow Diagram



6. CODE STRUCTURE (Template)

```
crypto-bot/
├─ config/
│   ├── api_keys.env (NOT in git!)
│   ├── coins.json (watchlist)
│   └─ parameters.json (strategy params)
├─ src/
│   ├── data_fetcher.py (Binance API calls)
│   ├── signals.py (momentum derivative logic)
│   ├── risk_manager.py (position sizing, exit rules)
│   ├── trader.py (execute orders)
│   ├── logger.py (track trades, stats)
│   └─ main.py (orchestrator - runs loop)
├─ backtest/
│   ├── backtest_engine.py
│   ├── historical_data.csv
│   └─ results/
├─ tests/
│   ├── test_signals.py
│   ├── test_risk.py
│   └─ test_integration.py
├─ alerts/
│   ├── telegram_notifier.py (optional)
│   └─ discord_webhook.py (optional)
└─ README.md
```

7. PARAMETER REFERENCE

Signal Parameters (Tune in backtest)

Parameter	Value	Range	Notes
Momentum lookback	5	3-9	Shorter = faster signal, more noise
Acceleration lookback	1	1-3	Usually fixed at 1

Parameter	Value	Range	Notes
4H ROC period	9-20	5-30	Trend strength
Volume SMA	20	14-50	Average volume baseline
Volume spike multiplier	1.2x	1.1-1.5x	How much > average

Risk Parameters (FIXED - don't change)

Parameter	Value	Why
TP %	2-3%	Realistic 1H move for alts
SL ATR multiple	1.5x	Dynamic, avoids whipsaws
Max candles hold	4-6	Small TF = quick in/out
Max positions	5-10	Correlation protection
Risk per trade	1-2%	Account preservation
Daily loss limit	5%	Circuit breaker

8. RISK MANAGEMENT

Position Sizing (Kelly Simplified)

```
python

def calculate_position_size(account_balance, win_rate, win_loss_ratio):
    """
    Kelly fraction = (win_rate * win_loss_ratio - (1 - win_rate)) / win_loss_ratio
    Conservative Kelly = Kelly fraction / 2.5 (reduce ruin risk)
    """
    kelly_fraction = (win_rate * win_loss_ratio - (1 - win_rate)) / win_loss_ratio
    position_size = account_balance * (kelly_fraction / 2.5)
    return min(position_size, account_balance * 0.02) # Cap at 2% per trade
```

Correlation Risk

DO NOT trade together:

- Multiple Layer2s in same candle (ARB + OP bleed together)
- All exchange tokens in 1H window (BNB + FTT + OKB)
- Meme cycle same time (DOGE + SHIB + PEPE)

Instead: Stagger entry 30min-1h apart for same category

Liquidation Risk (With Coinglass)

python

```
def filter_liquidation_risk(current_price, symbol, liquidation_data):
    """
    Reject entry if liquidation level too close
    """
    liq_level_long = liquidation_data['long']['level']
    liq_level_short = liquidation_data['short']['level']

    if current_price > liq_level_long * 0.98: # Within 2%
        return False # Avoid, whale trap likely

    return True
```

9. BACKTEST CHECKLIST

Before going LIVE:

- ☐ **Historical data:** 3-6 months minimum
- ☐ **Win rate:** Target 45-55% (quality > quantity)
- ☐ **Profit factor:** > 1.3 (avg win / avg loss)
- ☐ **Max drawdown:** < 12% account
- ☐ **Trades per month:** 20-40 (enough frequency)
- ☐ **Sharpe ratio:** > 1.0 (risk-adjusted return)
- ☐ **Parameter sensitivity:** Test ± 2 on all params (stable?)
- ☐ **Slippage test:** Include real Binance fees (0.1% taker)
- ☐ **Multiple market conditions:** Bull, bear, range tested
- ☐ **Correlation test:** Check across different coin groups

Example Backtest Metrics:

```
Total trades: 156
Win rate: 51.3%
Profit factor: 1.54
Avg win: +2.2%
Avg loss: -1.4%
Max drawdown: -8.7%
Total return: +47.2% (3 months)
Monthly avg: +15.7%
Sharpe ratio: 1.23
```

10. IMPLEMENTATION ROADMAP

Phase 1: Setup (Week 1)

- ☐ Create Binance API keys + test connection
- ☐ Code data fetcher (fetch 4H + 1H candles)
- ☐ Code signal calculator (momentum + acceleration)
- ☐ Set up logging/database

Phase 2: Backtest (Week 2-3)

- ☐ Gather 6-month historical data
- ☐ Code backtest engine
- ☐ Run 100+ trades backtest
- ☐ Optimize parameters
- ☐ Verify profitable edge

Phase 3: Paper Trade (Week 4)

- ☐ Run bot on paper trading (no real money)
- ☐ Monitor for 2-4 weeks
- ☐ Check slippage vs backtest
- ☐ Refine position sizing
- ☐ Test alerts/notifications

Phase 4: Live Micro (Week 5+)

- ☐ Go live with 0.1-0.5% account only
- ☐ Trade 2-4 weeks minimum
- ☐ Monitor daily P&L
- ☐ Check drawdown tolerance

- ☐ Scale if profitable + consistent

Phase 5: Full Scale (Month 2+)

- ☐ If 3+ weeks positive → scale to full size
 - ☐ Add Coinglass if liquidation losses occur
 - ☐ Monitor monthly, adjust parameters quarterly
 - ☐ Document all learnings
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11. MONITORING & ALERTS (Setup Now)

What to Monitor

- Daily P&L
- Win rate (rolling 20 trades)
- Max daily loss trigger
- Liquidation events (if applicable)
- API errors/downtime
- Unexpected drawdown

Alert Channels (Pick one)

```
python

# Option A: Telegram Bot (easiest)
telegram_webhook = "https://api.telegram.org/bot{TOKEN}/sendMessage"

# Option B: Discord Webhook
discord_webhook = "https://discord.com/api/webhooks/{ID}"

# Option C: Email (reliable)
SMTP server + Gmail credentials
```

Sample Alert

 BOT ALERT

Trade: ETHUSDT LONG
Entry: \$2,145.32
TP: \$2,204.86 (+2.8%)

SL: \$2,098.50 (-2.2%)

Bias: UP 

Signal: Acceleration building 

Volume spike: 1.4x 

Status:  EXECUTING

Time: 2024-01-15 14:23 UTC

12. PRODUCTION CHECKLIST

Before LIVE trading:

Security:

- ☐ API keys in environment variables (not hardcoded)
- ☐ Read-only initial testing key
- ☐ IP whitelist enabled on Binance
- ☐ Secret key never logged
- ☐ Bot runs on secure server (not public WiFi)

Infrastructure:

- ☐ Bot runs on 24/7 uptime (AWS EC2 / VPS)
- ☐ Error handling (crashes don't lose money)
- ☐ Database backup (daily)
- ☐ Monitoring (alert on bot crash)
- ☐ Rate limit handling (retry logic)

Money:

- ☐ Start with small account (\$500-1000)
 - ☐ Never trade borrowed money (no margin)
 - ☐ Daily loss limit: -5% max
 - ☐ Position size: 1-2% per trade max
 - ☐ Emergency stop-loss: Manual kill switch available
-

13. EXPECTED PERFORMANCE

Conservative Estimate (Post-Backtest)

Metric	Target	Realistic Range
Win Rate	50%	45-55%
Avg Win	+2.2%	+1.8% to +2.5%
Avg Loss	-1.4%	-1.2% to -1.8%
Profit Factor	1.5	1.3-1.8
Monthly Return	+10-15%	+5% to +20%
Drawdown Max	-8%	-5% to -12%

Reality check:

- Year 1 = learning + optimization phase
- Year 2+ = stable if parameters hold
- Market changes = need reoptimization every 3-6 months

14. TROUBLESHOOTING

Problem: Too many false signals

Solution:

- Increase volume spike threshold (1.2x → 1.5x)
- Add RSI filter (only if 40-60 range = unextreme)
- Reduce frequency (1H → 4H entry)

Problem: Losing on liquidation traps

Solution:

- Add Coinglass liquidation filter
- Check if funding rate extreme (>0.1%)
- Move SL above liquidation level

Problem: Slippage killing profits**Solution:**

- Use limit orders (not market)
- Increase TP % (3% → 4%)
- Trade larger cap coins only (SOL, AVAX, LINK)

Problem: Position correlation losses**Solution:**

- Reduce max concurrent positions (10 → 5)
- Add inter-trade delay (no 2+ entries same candle)
- Categorize coins by sector, limit per sector

Problem: Bot crashes/disconnects**Solution:**

- Add retry logic (3 attempts, 5sec wait)
 - Health check (restart if no response 10min)
 - Set up monitoring alerts
 - Run on VPS, not local computer
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15. QUICK REFERENCE CHECKLIST

Before Each Trading Day

- ☐ Bot is running (check logs)
- ☐ No errors in last 12h
- ☐ Account balance intact
- ☐ Daily P&L tracked
- ☐ Alerts working

Weekly (Every Sunday)

- ☐ Review all trades (win/loss analysis)
- ☐ Check win rate trend
- ☐ Verify parameters still optimal

- ☐ Backup database
- ☐ Any coins to add/remove from watchlist?

Monthly (1st of month)

- ☐ Full stats report
- ☐ Compare to backtest expectations
- ☐ Any parameter tweaks needed?
- ☐ Update coin watchlist
- ☐ Review market conditions (bull/bear/range)

Quarterly (Jan/Apr/Jul/Oct)

- ☐ Full reoptimization on new data
 - ☐ Test parameter sensitivity
 - ☐ Consider strategy iteration
 - ☐ Review P&L vs S&P 500 (benchmark)
-

16. USEFUL LINKS

Documentation:

- Binance API Docs: <https://binance-docs.github.io/apidocs/>
- python-binance: <https://python-binance.readthedocs.io/>
- Coinglass Docs: <https://docs.coinglass.com/>

Tools:

- Backtrader: <https://www.backtrader.com/>
- TradingView (chart validation): <https://www.tradingview.com/>
- Freqtrade (open-source bot): <https://www.freqtrade.io/>

Communities:

- r/algotrading
 - Crypto Discord communities (community-specific)
 - GitHub (search: crypto trading bot)
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17. FINAL NOTES

What This Strategy Is

- ✓ Momentum-based, not trend-following
- ✓ Small TF entries with large TF bias filter
- ✓ Risk-managed (fixed TP/SL)
- ✓ Backtest-validated
- ✓ Scalable from \$100 to \$100k+

What This Strategy Is NOT

- ✗ Get-rich-quick scheme
- ✗ Works in all market conditions (need regime detection)
- ✗ Guaranteed profitable (requires discipline)
- ✗ Set-and-forget (need monitoring)
- ✗ Holy Grail (will have losing periods)

Success Requirements

1. **Patience:** 2-4 weeks backtest before live
2. **Discipline:** Stick to rules, no emotional overrides
3. **Monitoring:** Daily check (15 min/day)
4. **Flexibility:** Adapt to market regime changes
5. **Risk management:** Never risk more than planned

Version: 1.0 **Last Updated:** January 2025 **Status:** Ready for implementation